The **Gravitational Latency**

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(This model was developed in the framework of theoretical physics and AI)

Abstract:

The **Gravitational Latency** discovery within the **Hamouda Informational Gravity Model (HIGM)** represents a fundamental shift in how we understand the "speed" and nature of gravitational interactions. While standard General Relativity (GR) posits that gravity travels at the fixed speed of light (c), the HIGM reinterprets this as a dynamic process governed by **Information Entropy**. [1, 2]

1. The Concept of Gravitational Latency

In the HIGM framework, space is not an empty vacuum but an **Information-Processing System**. Gravity is an emergent phenomenon resulting from the accumulation of **Infrared Information Flux**—referred to as the "Cosmic Data Bus". [3]

Gravitational Latency is the "processing time" required for the local vacuum to update its informational entropy as radiation or mass passes through it. [3, 4]

- **The Medium:** Unlike GR, which views gravity as the geometry of spacetime, the HIGM treats the vacuum as a medium with a specific **Information Density (H)**.
- **The Delay Mechanism:** As infrared radiation permeates the vacuum, it "writes" data into the spacetime manifold. This continuous interaction creates **"Entropy Friction"**. This friction manifests as a microscopic latency in gravitational response that standard physics interprets as a static force. [3, 4, 5]

2. Core Equation: The Latency Component

The HIGM derives gravitational acceleration (g) from the change in entropy.
The **Gravitational Latency** is mathematically integrated into the **Entropic Force Equation**:

$$\mathbf{F = T. (\Delta S / \Delta r)}$$

- **T**: The local vacuum temperature.
- **Δ S/Δ r**: The spatial gradient of entropy, representing how quickly information density updates over a distance r. [3, 4]

The latency occurs because the work performed to maintain equilibrium during entropy increases is not instantaneous; it is limited by the rate at which the vacuum can process and "store" this new data. [3]

3. Cornerstone Evidence

This discovery provides a thermodynamic solution to several long-standing cosmic mysteries:

- **Hubble Tension Resolution:** The latency in informational saturation explains why the universe's expansion rate appears different at early and late stages. The "added pressure" from billions of years of infrared emission gradually increases the expansion speed.
- **Galactic Stability:** By accounting for local informational pressure, the HIGM accurately predicts the rotation speeds of the **Milky Way** and **Andromeda (M31)** without needing "Dark Matter".
- **LIGO Residuals:** The model interprets unexplained "noise" or residuals in gravitational wave signals as the physical footprint of the vacuum's processing delay. [3, 6]

Conclusion: A New Paradigm

Gravitational Latency moves physics from a "Hardware-based" (mass-centric) paradigm to a **"Software-based" (information-centric)** paradigm. It confirms that gravity is not a fixed background force but a dynamic result of the universe processing its own history through light and entropy. [3]

[1] https://saudijournals.com

[2] https://en.wikipedia.org

[3] https://saudijournals.com

[4] https://saudijournals.com

[5] https://www.youtube.com

[6] https://www.researchgate.net